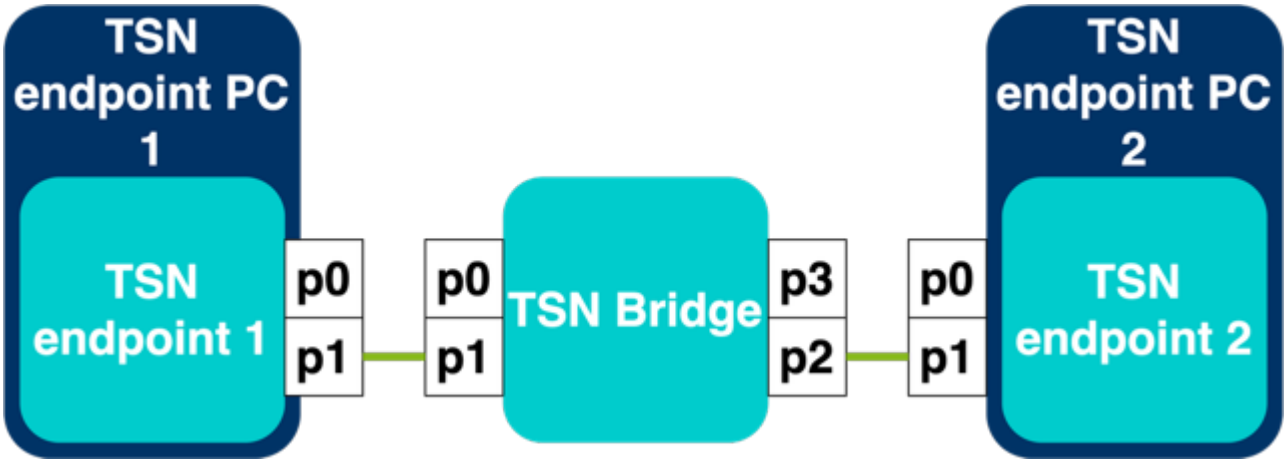
TSN

TSN capability has been successfully adddes to the Trial Network macimaci

Time-Sentive Networking Trial Network

The Time-Sentive Networking (TSN) Trial Network (TN) architecture is shown below, in which 2 TSN endpoints are interconnected through a TSN bridge.



TSN configuration

The TSN configuration can be performed via SSH (cli) using a terminal or dashboard (gui) using a browser. Details on how to configure the different TSN features can be found in the TSN configuration guide. For example, the following command is used to configure the Time Aware Shaper (IEEE 802.1Qbv):

```
spt_config_reg -n 0 -w qbv_config_file.json
```

In this case, the specific parameters are defined in qbv_config_file.json and are applied only on port 0 of the device (indicated by -n 0). The following image shows the TSN configuration defined in qbv_config_file.json.

192.168.2.100/?cfg=tas&attr=adv_conf

TIME AWARE SHAPER | Administrative

	Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
Gate states	☑	☑	☑	☑	☑	☑	☑	☑

Cycle time (ns)	60000000
Cycle time extension (ns)	1000
Base time seconds	0
Base time nanoseconds	0
Control list length	6

Time interval (ns)	Slot/Queue	Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
10000000	Slot 0	☐	☐	☑	☐	☐	☐	☐	☐
10000000	Slot 1	☐	☐	☐	☐	☐	☐	☐	☑
10000000	Slot 2	☐	☐	☐	☐	☑	☐	☐	☐
10000000	Slot 3	☐	☐	☐	☐	☐	☑	☐	☐
10000000	Slot 4	☑	☐	☐	☐	☐	☐	☐	☐
10000000	Slot 5	☐	☑	☐	☑	☐	☐	☑	☐

The TSN configuration can also be checked using the command:

```
spt_qbv_config -n 0 -r
```

Note that only the administrative values are shown. However, the operative values will be the same and are also shown in the command output:

```
-------------------------------------------------------------
Port ID: 0
Gate Enabled: 0

---------------------
Administrative values
---------------------
Gate States: 255
Control List Length: 6

        ------------
        Control List
        ------------

        Index: 0
        Operation Name: set-gate-states
        Gate States Value: 32
        Time Interval Value: 10000000

        Index: 1
        Operation Name: set-gate-states
        Gate States Value: 1
        Time Interval Value: 10000000

        Index: 2
        Operation Name: set-gate-states
        Gate States Value: 8
        Time Interval Value: 10000000

        Index: 3
        Operation Name: set-gate-states
        Gate States Value: 4
        Time Interval Value: 10000000

        Index: 4
        Operation Name: set-gate-states
        Gate States Value: 128
        Time Interval Value: 10000000

        Index: 5
        Operation Name: set-gate-states
        Gate States Value: 82
        Time Interval Value: 10000000
```

```
Cycle Time Numerator: 60000000
Cycle Time Denominator: 1000000000
Cycle Time Extension: 1000
Base Time Seconds: 0
Base Time Nanoseconds: 0
---------------------
```

Login Credentials

Information required to access the TSN components of the trial network.

Device	IP address	User	Password	Connection
TSN endpoint 1 PC	X.X.X.X	tsnep1	tsnep1	SSH or VNC
TSN endpoint 2 PC	X.X.X.X	tsnep2	tsnep2	SSH or VNC
TSN endpoint 1	192.168.2.100	relyum	relyum	SSH
TSN endpoint 1	192.168.2.100	admin	relyum	Dashboard
TSN endpoint 2	192.168.2.200	relyum	relyum	SSH
TSN endpoint 2	192.168.2.200	admin	relyum	Dashboard
TSN bridge	192.168.2.10	relyum	relyum	SSH
TSN bridge	192.168.2.10	relyum	admin	Dashboard

ANNEX A: Relyum TSN bridge CLI configuration

1. CLI configuration tool

1.1 Design functionalities

These are the listed functionalities by default in the -TSN-PCIe :

- reg: shows the raw registers map (32-bit values).
- core info: shows version information and HW/SW licensed features.
- ptp: allows to configure the PTP functionality of the IP core.

- switching: shows configuration _elds related to switching parameters.
- rstp: shows configuration _elds for rstp protocol.
- mac query: shows the learned MAC addresses.
- tsn: allows to configure the TSN parameters of the device.
- port config: shows configuration _elds particular to each Ethernet port and non
- related to other functionalities.
- stats: shows the traffic statistics of each Ethernet port.
- serv: allows controlling the services available in the device.

Examples:

- switching
  - mes_core_version (reg: 0x00000500 width: 32 offset: 0)
  - mes_switching_mask_control (reg: 0x00000504 width: 32 offset: 0)
  - mes_switching_mask (reg: 0x00000508 width: 32 offset: 0)
  - mes_learning_enable (reg: 0x00000600 width: 1 offset: 0)
  - mes_ageing_enable (reg: 0x00000604 width: 1 offset: 0)
  - mes_ageing_time (reg: 0x00000608 width: 20 offset: 0)
  - mes_clear_dynamic_mac_entries (reg: 0x0000060C width: 1 offset: 0)
  - mes_clear_static_mac_entries (reg: 0x0000060C width: 1 offset: 1)
  - mes_clear_complete_mac_entries (reg: 0x0000060C width: 1 offset: 2)
  - mes_clear_vlan_id_mac_entries (reg: 0x0000060C width: 1 offset: 3)
  - mes_clear_vlan_id_entry (reg: 0x0000060C width: 12 offset: 30)
  - mes_static_mac_address_control (reg: 0x00000610 width: 32 offset: 0)
  - mes_static_mac_address_lo (reg: 0x00000614 width: 32 offset: 0)
  - mes_static_mac_address_hi_vlan_id (reg: 0x00000618 width: 32 offset: 0)
  - mes_static_mac_address_source_ports (reg: 0x0000061C width: 32 offset: 0)
  - mes_global_port_based_vlan_control (reg: 0x00000700 width: 1 offset: 0)
  - mes_global_reset_vlan_configuration (reg: 0x00000704 width: 1 offset: 0)
  - mes_global_allowed_vlan (reg: 0x00000708 width: 12 offset: 0)
  - mes_ethertype_custom_s_port (reg: 0x0000070C width: 16 offset: 0)
  - mes_vlan_port_id (reg: 0x00000710 width: 8 offset: 0)
  - mes_vlan_port_mode (reg: 0x00000714 width: 2 offset: 0)
  - mes_vlan_port_type (reg: 0x00000718 width: 2 offset: 0)
  - mes_port_vlan (reg: 0x0000071C width: 16 offset: 0)
  - mes_vlan_ingress_filtering (reg: 0x00000720 width: 1 offset: 0)
  - mes_vlan_ingress_acceptance (reg: 0x00000724 width: 2 offset: 0)
  - mes_port_vlan_egress_tagging (reg: 0x00000728 width: 2 offset: 0)
  - mes_port_vlan_priority_tagged_egress_tagging (reg: 0x00000728 width: 2 offset: 4)

    - mes_vlan_id_configuration (reg: 0x0000072C width: 12 offset: 0)
    - mes_vlan_id_member_configuration (reg: 0x00000730 width: 1 offset: 0)
    - mes_vlan_id_member_ports_0 (reg: 0x00000734 width: 32 offset: 0)
    - mes_custom_vlan_untag_configuration (reg: 0x0000073C width: 1 offset: 0)
    - mes_custom_vlan_untag_mask_0 (reg: 0x00000740 width: 32 offset: 0)
    - preemption_port_id (reg: 0x00001100 width: 8 offset: 0)
    - preemption_support (reg: 0x00001104 width: 1 offset: 0)
    - preemption_enable (reg: 0x00001108 width: 1 offset: 0)
    - preemption_hold_release_enable (reg: 0x0000110C width: 1 offset: 0)
    - preemption_priority_queues (reg: 0x00001110 width: 32 offset: 0)
    - host_access_control_enable (reg: 0x00001200 width: 1 offset: 0)
    - host_access_control_write (reg: 0x00001204 width: 1 offset: 0)
    - host_access_control_clear (reg: 0x00001204 width: 1 offset: 1)
    - host_access_control_index (reg: 0x00001208 width: 4 offset: 0)
    - host_access_control_mac_addr_lo (reg: 0x0000120C width: 32 offset: 0)
    - host_access_control_mac_addr_hi (reg: 0x00001210 width: 16 offset: 0)
    - host_access_control_ethertype (reg: 0x00001214 width: 16 offset: 0)
    - host_access_control_mask (reg: 0x00001218 width: 2 offset: 0)
- tsn
    - tsn_qbv_port_id (reg: 0x00002000 width: 8 offset: 0)
    - tsn_qbv_gate_enabled (reg: 0x00002004 width: 1 offset: 0)
    - tsn_qbv_admin_gate_states (reg: 0x00002008 width: 8 offset: 0)
    - tsn_qbv_oper_gate_states (reg: 0x0000200C width: 8 offset: 0)
    - tsn_qbv_admin_control_list_length (reg: 0x00002010 width: 32 offset: 0)
    - tsn_qbv_oper_control_list_length (reg: 0x00002014 width: 32 offset: 0)
    - tsn_qbv_admin_list_pointer (reg: 0x00002018 width: 32 offset: 0)
    - tsn_qbv_admin_gate_state_values (reg: 0x0000201C width: 10 offset: 0)
    - tsn_qbv_admin_time_interval (reg: 0x00002020 width: 32 offset: 0)
    - tsn_qbv_oper_list_pointer (reg: 0x00002024 width: 32 offset: 0)
    - tsn_qbv_oper_gate_state_values (reg: 0x00002028 width: 10 offset: 0)
    - tsn_qbv_oper_time_interval (reg: 0x0000202C width: 32 offset: 0)
    - tsn_qbv_admin_cycle_time (reg: 0x00002030 width: 32 offset: 0)
    - tsn_qbv_oper_cycle_time (reg: 0x00002034 width: 32 offset: 0)
    - tsn_qbv_admin_cycle_time_extension (reg: 0x00002038 width: 32 offset: 0)
    - tsn_qbv_oper_cycle_time_extension (reg: 0x0000203C width: 32 offset: 0)
    - tsn_qbv_admin_base_time_seconds_31_0 (reg: 0x00002044 width: 32 offset: 0)
    - tsn_qbv_admin_base_time_nanoseconds (reg: 0x00002048 width: 32 offset: 0)
    - tsn_qbv_oper_base_time_seconds_31_0 (reg: 0x00002050 width: 32 offset: 0)

- tsn_qbv_oper_base_time_nanoseconds (reg: 0x00002054 width: 32 offset: 0)
- tsn_qbv_configuration_change_request (reg: 0x00002058 width: 32 offset: 0)
- tsn_qbv_configuration_change (reg: 0x0000205C width: 1 offset: 0)
- tsn_qbv_configuration_change_time_seconds_31_0 (reg: 0x00002064 width: 32 offset: 0)
- tsn_qbv_configuration_change_time_nanoseconds (reg: 0x00002068 width: 32 offset: 0)
- tsn_qbv_configuration_apply_request (reg: 0x0000206C width: 32 offset: 0)
- tsn_qbv_configuration_apply (reg: 0x00002070 width: 1 offset: 0)
- tsn_qbv_configuration_apply_time_seconds_31_0 (reg: 0x00002078 width: 32 offset: 0)
- tsn_qbv_configuration_apply_time_nanoseconds (reg: 0x0000207C width: 32 offset: 0)
- tsn_qbv_configuration_pending (reg: 0x00002080 width: 1 offset: 0)
- tsn_qbv_supported_list_max (reg: 0x00002084 width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_0 (reg: 0x00002088 width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_1 (reg: 0x0000208C width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_2 (reg: 0x00002090 width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_3 (reg: 0x00002094 width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_4 (reg: 0x00002098 width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_5 (reg: 0x0000209C width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_6 (reg: 0x000020A0 width: 32 offset: 0)
- tsn_qbv_queuemaxsdu_priority_queue_7 (reg: 0x000020A4 width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_0 (reg: 0x000020A8 width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_1 (reg: 0x000020AC width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_2 (reg: 0x000020B0 width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_3 (reg: 0x000020B4 width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_4 (reg: 0x000020B8 width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_5 (reg: 0x000020BC width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_6 (reg: 0x000020C0 width: 32 offset: 0)
- tsn_qbv_transmission_overrun_queue_7 (reg: 0x000020C4 width: 32 offset: 0)
- tsn_qbv_tx_logic_latency_1000Mbps (reg: 0x000020C8 width: 32 offset: 0)
- tsn_qbv_tx_logic_latency_100Mbps (reg: 0x000020CC width: 32 offset: 0)
- tsn_qbv_tx_logic_latency_10Mbps (reg: 0x000020D0 width: 32 offset: 0)
- tsn_allowed_gates (reg: 0x000020D4 width: 8 offset: 0)
- tsn_configuration_change_past (reg: 0x000020D8 width: 32 offset: 0)
- tsn_cbs_port_id (reg: 0x00002100 width: 8 offset: 0)
- tsn_cbs_priority_buffer (reg: 0x00002104 width: 3 offset: 0)
- tsn_cbs_enable (reg: 0x00002108 width: 1 offset: 0)
- tsn_cbs_queue_idle_slope (reg: 0x0000210C width: 16 offset: 0)
- tsn_stream_identification_write (reg: 0x00002200 width: 1 offset: 0)
- tsn_stream_identification_read (reg: 0x00002200 width: 1 offset: 1)

    - tsn_stream_identification_clear_stream (reg: 0x00002200 width: 1 offset: 2)
    - tsn_stream_identification_clear_all_streams (reg: 0x00002200 width: 1 offset: 3)
    - tsn_stream_identification_use_priority (reg: 0x00002200 width: 1 offset: 8)
    - tsn_stream_id_mac_lo (reg: 0x00002204 width: 32 offset: 0)
    - tsn_stream_id_mac_hi (reg: 0x00002208 width: 16 offset: 0)
    - tsn_stream_id_vlan (reg: 0x0000220C width: 32 offset: 0)
    - tsn_stream_handle (reg: 0x00002210 width: 32 offset: 0)
    - tsn_stream_param_control (reg: 0x00002300 width: 32 offset: 0)
    - tsn_stream_param_stream_handle (reg: 0x00002304 width: 32 offset: 0)
    - tsn_max_frame_size (reg: 0x00002330 width: 32 offset: 0)
    - tsn_ipv (reg: 0x00002334 width: 32 offset: 0)
    - tsn_meter_id (reg: 0x0000233C width: 32 offset: 0)
    - tsn_metering_mode (reg: 0x00002340 width: 5 offset: 0)
    - tsn_cir (reg: 0x00002344 width: 32 offset: 0)
    - tsn_eir (reg: 0x00002348 width: 32 offset: 0)
    - tsn_cbs (reg: 0x0000234C width: 32 offset: 0)
    - tsn_ebs (reg: 0x00002350 width: 32 offset: 0)
    - tsn_qci_port (reg: 0x00002400 width: 8 offset: 0)
    - tsn_psfp_en (reg: 0x00002404 width: 1 offset: 0)
    - tsn_cb_proxy_port (reg: 0x00002500 width: 32 offset: 0)
    - tsn_cb_stream_handler_reset (reg: 0x00002508 width: 32 offset: 0)
    - tsn_cb_recovery_reset (reg: 0x0000250C width: 1 offset: 0)
    - tsn_cb_recovery_enable (reg: 0x00002510 width: 1 offset: 0)
    - tsn_cb_recovery_mode (reg: 0x00002514 width: 1 offset: 0)
    - tsn_cb_recovery_history (reg: 0x00002518 width: 4 offset: 0)
    - tsn_cb_recovery_reset_time (reg: 0x0000251C width: 32 offset: 0)
    - tsn_cb_recovery_rx_take_no_sequence (reg: 0x00002520 width: 1 offset: 0)
    - tsn_cb_recovery_tx_take_no_sequence (reg: 0x00002520 width: 1 offset: 1)
    - tsn_active_stream_translation_control (reg: 0x00002600 width: 32 offset: 0)
    - tsn_active_stream_translation_port_id (reg: 0x00002604 width: 8 offset: 0)
    - tsn_active_stream_translation_stream_handle (reg: 0x00002608 width: 32 offset: 0)
    - tsn_active_stream_translation_mac_lo (reg: 0x0000260C width: 32 offset: 0)
    - tsn_active_stream_translation_mac_hi (reg: 0x00002610 width: 16 offset: 0)
    - tsn_active_stream_translation_vlan (reg: 0x00002614 width: 32 offset: 0)
- port_config
    - half_duplex (reg: 0x00000018 width: 1 offset: 0)
    - enable_port (reg: 0x00000068 width: 1 offset: 0)
- stats (available on each PORT)

- phy_speed (reg: 0x0000001C width: 2 offset: 0)
- rx_frames (reg: 0x00000020 width: 32 offset: 0)
- tx_frames (reg: 0x00000024 width: 32 offset: 0)
- crc_err_frames (reg: 0x00000028 width: 32 offset: 0)
- lanid_err_frames (reg: 0x0000002C width: 32 offset: 0)
- reset_all_cnt (reg: 0x0000003C width: 1 offset: 0)
- en_stat_count (reg: 0x00000040 width: 1 offset: 0)
- rx_drop_overflow_frame (reg: 0x00000044 width: 32 offset: 0)
- rx_unicast_frame (reg: 0x00000048 width: 32 offset: 0)
- rx_multicast_frame (reg: 0x0000004C width: 32 offset: 0)
- rx_broadcast_frame (reg: 0x00000050 width: 32 offset: 0)
- rx_vlan_tag_frame (reg: 0x00000054 width: 32 offset: 0)
- rx_ptp_frame (reg: 0x00000058 width: 32 offset: 0)
- rx_overlength_frame (reg: 0x0000005C width: 32 offset: 0)
- rx_underlength_frame (reg: 0x00000060 width: 32 offset: 0)
- rx_data_bytes (reg: 0x00000064 width: 32 offset: 0)
- vlan_id_stat (reg: 0x0000006C width: 12 offset: 0)
- vlan_stat_en (reg: 0x0000006C width: 1 offset: 12)
- tx_drop_overflow_frame (reg: 0x00000070 width: 32 offset: 0)
- tx_unicast_frame (reg: 0x00000074 width: 32 offset: 0)
- tx_multicast_frame (reg: 0x00000078 width: 32 offset: 0)
- tx_broadcast_frame (reg: 0x0000007C width: 32 offset: 0)
- tx_vlan_tag_frame (reg: 0x00000080 width: 32 offset: 0)
- tx_ptp_frame (reg: 0x00000084 width: 32 offset: 0)
- tx_data_bytes (reg: 0x00000088 width: 32 offset: 0)
- rx_preemption_start_frame (reg: 0x000000C0 width: 32 offset: 0)
- rx_preemption_continue_frame (reg: 0x000000C4 width: 32 offset: 0)
- rx_preemption_bad_sequence_continue_frame (reg: 0x000000C8 width: 32 offset: 0)
- rx_preemption_crc_error_frame (reg: 0x000000CC width: 32 offset: 0)
- tx_preemption_start_frame (reg: 0x000000D0 width: 32 offset: 0)
- tx_preemption_continue_frame (reg: 0x000000D4 width: 32 offset: 0)

1.2. Read operations

```
$ spt_config_reg -f CORE:field_name -r

Example:

$ spt_config_reg -f INTERNAL_PORT:tx_frames -r Read value: 42465 (HEX: 0x0000A5E1)
```

1.3. Write operations

```
$ spt_config_reg -f CORE:field_name -w value

Example:

$ spt_config_reg -f SWITCH:tsn_cbs -w 1 Operation successful
```

2. QBV config

2.1. Read current QBV config

```
$ spt_qbv_config -r

Operation completed

Status code: SUCCESS

------------------------------------------------------------
Port ID: 0
Gate Enabled: 0

---------------------
Administrative values
---------------------

Gate States: 255
Control List Length: 2

------------
Control List
------------

  Index: 0

  Operation Name: set-gate-states
  Gate States Value: 240
  Time Interval Value: 50000

  Index: 1

  Operation Name: set-gate-states Gate
  States Value: 15
  Time Interval Value: 50000

Cycle Time Numerator: 1
Cycle Time Denominator: 1000
Cycle Time Extension: 1000
Base Time Seconds: 0

Base Time Nanoseconds: 0
---------------------

---------------------
Operative values
```

```
---------------------

Gate States: 255
Control List Length: 0

------------
Control List
------------

Cycle Time Numerator: 4294967295
Cycle Time Denominator: 1000000000
Cycle Time Extension: 4294967295
Base Time Seconds: 0
Base Time Nanoseconds: 0

---------------------

Config Change: 0
Config Change Time Seconds: 0
Config Change Time Nanoseconds: 0 Tick Granularity: 9
Current Time Seconds: 3022709 Current Time Nanoseconds: 923154673 Config Pending: 0
Config Change Error: 0
Suported List Max: 256
```

2.2. Write QBV config

```
$ spt_qbv_config -w qbv_config_file_test.json
```

qbv_config_file_test.json

```
{
   "qbv_gate_parameters":{
      "gate_enabled":false,
      "admin":{
         "gate_states":255,
         "control_list_length":2,
         "control_list":[
            {
               "index":0,
               "operation_name":"set-gate-states",
               "gate_states_value":240,
               "time_interval_value":50000
            },
            {
               "index":1,
               "operation_name":"set-gate-states",
               "gate_states_value":15,
               "time_interval_value":50000
            }
         ],
         "cycle_time":{
            "numerator":1,
            "denominator":1000
         },
         "cycle_time_extension":1000,
         "base_time":{
            "seconds":0,
            "nanoseconds":0
         }
      },
      "config\_change":false
   }
}
```


3. QAV config

3.1. Read current QAV config

```
$ spt_qav_config -r

------------------------------------------------------------

Priority queue: 0

Idle slope: 100

CBS enable: 0
------------------------------------------------------------
```

3.2. Write QAV config

```
$ spt_qav_config -w qav_config_file_test.json
```

qav_config_file_test.json

```
{
  "cbs_params" : {
    "cbs_enable": false,
    "idle_slope": 100
  }
}
```

ANNEX B: qvb_config_file.json


```
{
    "qbv_gate_parameters": {
        "gate_enabled": true,
        "admin": {
            "gate_states": 255,
            "control_list_length": 6,
            "control_list": [
                {
                    "index": 0,
                    "operation_name": "set-gate-states",
                    "gate_states_value": 32,
                    "time_interval_value": 10000000
                },
                {
                    "index": 1,
                    "operation_name": "set-gate-states",
                    "gate_states_value": 1,
                    "time_interval_value": 10000000
                },
                {
                    "index": 2,
                    "operation_name": "set-gate-states",
                    "gate_states_value": 8,
                    "time_interval_value": 10000000
                },
                {
                    "index": 3,
                    "operation_name": "set-gate-states",
                    "gate_states_value": 4,
                    "time_interval_value": 10000000
                },
                {
                    "index": 4,
                    "operation_name": "set-gate-states",
                    "gate_states_value": 128,
                    "time_interval_value": 10000000
                },
                {
                    "index": 5,
                    "operation_name": "set-gate-states",
```

```
                "gate_states_value": 82,
                "time_interval_value": 10000000
            }
        ],
        "cycle_time": {
            "numerator": 60000000,
            "denominator": 1000000000
        },
        "cycle_time_extension": 1000,
        "base_time": {
            "seconds": 0,
            "nanoseconds": 0
        }
    },
    "config_change": true
}
}
```